

Social Centrality Differentiation Shapes the Dynamics of Linguistic Diffusion

Abstract

The diffusion of linguistic innovations depends on how speakers are embedded in their social networks. While central individuals are often thought to facilitate spread, their precise contribution remains uncertain. This study examines how the number of social ties interacts with communication scenarios to shape diffusion outcomes within a simplified computational model. Using agent-based simulations and an equation-based model, we evaluate how network position predicts the success rate, timing, and trajectory of innovation spread. Within this model, results indicate that, when individuals with more connections are assigned as the initial sources of innovations, the innovations are less likely to become successful and tend to show lower diffusion intensity and lower final diffusion levels. Under the modeled interaction rules, one-to-one interactions appear to promote stable adoption, whereas one-to-many settings favor rapid dissemination. These findings identify mechanisms and conditions under which these diffusion patterns emerge in the model: peripheral positions may provide more favorable starting points for successful diffusion, central adopters may encounter innovations earlier, and the distribution of social ties is a key factor linking both processes. They should therefore be interpreted as model-contingent results rather than general conclusions about linguistic innovation behavior.

Keywords linguistic diffusion; social networks; agent-based modelling; social centrality; language change

Introduction

Diffusion, as a universal phenomenon in various biological systems, has long been studied in terms of why and how it appears, succeeds, and fails (Newman, 2003; Rogers, 2003; Centola and Macy, 2011). Language diffusion has likewise attracted sustained attention from researchers of language change (Shen, 1990; Zhang and Gong, 2013; Blythe and Croft, 2012). In the language domain, diffusion has at least three senses: geographically, it refers to the process whereby a variant distributes more widely in real-world space (e.g., grammatical construction variants in East Anglia, see Trudgill (1974)); socially, it refers to the process whereby a language or a variant becomes popular in a community (e.g., different pronunciations of /r/ for different social classes, see Labov (2001)); internally, it refers to the process whereby a variant competes with others (e.g., gradual sound change in different lexicon groups, see Wang (1969)). To sum up, linguistic diffusion explores how the proportion of linguistic variants or their users changes over time. However, why it is patterned the way it is and what factors govern its dynamics remain unresolved issues.

If language emerged as a system that accommodates changes in social structure (Friedrich, 1966; Weinreich et al., 1968), its patterns are bound to be affected by structural properties and interactional mechanisms of society. As one of the crucial factors, the social centrality of speakers plays an important role in linguistic diffusion (Ke et al., 2008; Trudgill, 2011; Wray and Grace, 2007).

Different individuals play different roles in investigating patterns of language change due to their central or peripheral position in social networks (Blythe and Croft, 2021; Fagyal

et al., 2010; Labov, 1972). On one hand, individuals with higher centrality were seen as drivers or leaders for linguistic changes (Labov, 2001). Specifically, changes in progress are more easily observed in their speech (Eckert, 1989). On the other hand, those with lower centrality tend to innovate or introduce new variants and recycle old ones (Milroy and Milroy, 1985; Trudgill, 1999). In addition, these structural properties may interact with social factors, such as class, sex, and age (Chambers, 2009; Sharma and Dodsworth, 2020).

Though potential correlations between individual centrality and linguistic change have been supported by some evidence, important debates remain. Details on the effect of centrality on diffusion dynamics are limited to a qualitative description, rather than a quantitative comparison (Ke et al., 2008; Fagyal et al., 2010). In particular, how centrality contributes to the properties of diffusion curves, which often exhibit a sigma pattern (Blythe and Croft, 2012; Labov, 2007), remains unknown. Furthermore, emphasis on different aspects of social connections, such as their numbers, weights, directions, and homogeneity, might lead to distinct explanations for changes. For example, the weak-tie model suggested that individuals with more weak ties might be necessary for diffusion (Granovetter, 1973; 1983), which helps explain the changes led by females in Belfast (Milroy and Milroy, 1978). Moreover, homophily, which refers to the proportion of ties with the same labels (primarily ethnic) in a network, can also provide explanations for the maintenance of vernacular linguistic forms (Hoffman and Walker, 2010; Sharma and Sankaran, 2011). Moreover, it remains unclear whether the role of individuals differs across various interaction mechanisms.

Thus, it is difficult to establish reliable connections between aspects of centrality and diffusion dynamics.

Although previous works tried to isolate centrality effects from non-network factors (Labov, 2001), they failed to disentangle aspects of centrality that are sensitive to change from others. Others exploring the relationship between centrality and diffusion have yielded conflicting results: some endorse the two-step flow of influence model (which suggests central nodes drive spread, model details see (Lazarsfeld et al., 1949); and examples see Eckert (1989; 2000)), while others support the weak-tie model (peripheral nodes play a key role, Milroy and Milroy (1992)).

Here, we examine the effect of centrality based on a simplified measure and show how it shapes aspects of linguistic dynamics in the diffusion process under different scenarios. Differences in the degree of their connection with networks shape individual roles. Individuals with higher centrality are often assumed to drive linguistic change, while those with lower centrality are often associated with the introduction of new variants (Fagyal et al., 2010). From a quantitative perspective, we demonstrate how centrality influences the diffusion process and its dynamics through computer simulation. That is, more central adopters tend to adopt new variants earlier, while diffusions originating from more central individuals show scenario-dependent timing, lower success rates, lower diffusion intensity, and lower final diffusion levels.

Up to now, few studies have examined connections between aspects of centrality and detailed properties of diffusion dynamics, although numerous computational simulation studies have attempted to do so (Fagyal et al., 2010; Swarup et al., 2011; Josserand et al., 2024). We provide new insights through agent-based models and mathematical models that can generate and analyze diffusion curves affected by centrality. Here, we focus on the following questions: (1) How does centrality influence quantitative properties (i.e., success rate of diffusion, adoption time of individuals, shape of diffusion curves) of the diffusion process... (2) Does this effect vary in different scenarios...

Methods

We combine agent-based and equation-based models to examine the role of individuals with varying centrality under two scenarios. The agent-based model is employed to simulate the diffusion process influenced by centrality, scenarios, and other fundamental parameters. In contrast, the equation-based model is used to quantify the characteristics of diffusion dynamics. These two complementary approaches help establish an accurate link between factors and linguistic dynamics (Gong et al., 2014).

Agent-based models

Social structures

Language is a human-driven complex adaptive system (Liu, 2018), and its change always appears in a speech community where individuals exhibit diverse social characteristics. Real social interactions often do not follow the uniform mixing assumption. In contrast to approaches that rely on attributes such as sex, class, or other individual traits, social networks provide a more streamlined method for modeling the structure

of speech communities by mapping interconnections among individuals (Milroy and Milroy, 1985; Labov, 2001). In addition, dynamic social interactions reflected by networks might be closer to real situations than fixed approaches (Milroy and Llamas, 2013). With the above advantages, social network analysis has been widely applied to investigate the role of centrality (Labov, 2001; Milroy and Milroy, 1985; Trudgill, 1974). Hence, network-based structures are chosen to spread variants.

In a social network, an individual can spread variants to their neighbors only. Different network structures can lead to varying outcomes (Ke et al., 2008; Gong et al., 2008; Josserand et al., 2024; Raviv et al., 2019; 2020). Real social networks often exhibit small-world properties and power-law degree distributions (Watts and Strogatz, 1998; Barabási and Albert, 1999), where individuals diversify into either centers or peripheries. As a simplified implementation of scale-free networks, we use the Barabasi-Albert algorithm ($n=500$, $m=2$, where n is the network size, and m is the number of edges linked at one tick, see Barabási and Albert (1999)) to generate undirected and unweighted networks in which individuals interact.

Agents

Linguistic variants and users have commonly been employed as agents in previous models (Gong et al., 2014). In our study, we focus on the diffusion of linguistic variants within social networks, and thus, users are designated as agents. These agents will select linguistic variants with probabilities contingent on the functions or social impact of those forms (Nettle, 1999; Croft, 2003; Kirby, 1999), with the latter being particularly influential. This selection pressure consists of overt and covert prestige (Labov, 2001; Trudgill, 1972). Linguistic forms associated with overt prestige tend to gain widespread popularity among the general population, whereas specific social groups typically embrace those with covert prestige. For example, people in Martha's Vineyard resist the high-prestige pronunciation of /ay/ and /aw/ strongly (Labov, 1963).

To simulate prestige-based selection, agents in our model are divided into two types. Agents who prefer overt prestige (AO) are more likely to adopt a high-prestige option, while those favoring covert prestige (AC) are inclined to oppose its dissemination. We use the parameter *PropAO* to specify the proportion of AO agents, with AC agents constituting the remaining proportion, $1 - \text{PropAO}$. The rate of AO adopting the high-prestige variant is doubled, whereas that for AC is halved.

In addition, there often exists a threshold problem in diffusion, that is, an innovation's perceived advantage must be substantial enough to facilitate widespread adoption across a population (Nettle, 1999). Previous models have suggested that high prestige bias is necessary for diffusion (Abrams and Strogatz, 2003; Ke et al., 2008; Blythe and Croft, 2012). Our model adopts the parameter *PreBias* to modify the threshold.

Therefore, AO adopts the new variant with probability $2r\text{PreBias}$ and the old variant with $0.5r$. In contrast, AC adopts the new variant with probability $0.5r\text{PreBias}$ and the old variant with $2r$, where $r = 0.1$ is the fixed basic adoption rate. Because these expressions may exceed 1 under some parameter settings, the implemented adoption probability is bounded at 1 and calculated as $\min(1, p)$, where p is the value given by the corresponding expression. Table 1 reports

Table 1. Effective adoption probabilities after bounding the calculated probabilities at 1.

Agent type	Share	Variant	$PreBias = 10$	$PreBias = 20$
AO	$PropAO$	New	1.00	1.00
AO	$PropAO$	Old	0.05	0.05
AC	$1 - PropAO$	New	0.50	1.00
AC	$1 - PropAO$	Old	0.20	0.20

the resulting effective probabilities. With the parameter values examined here, the probability that an AO agent adopts the new variant is already capped at 1 for both levels of $PreBias$, whereas the corresponding probability for an AC agent increases from 0.5 to 1. The probabilities of adopting the old variant remain unchanged. Thus, within the tested range, $PreBias$ has a type-dependent and bounded effect rather than producing a uniform increase in prestige bias across all agents. Because $PropAO$ ranges from 0 to 0.6, AC agents constitute 40%–100% of the population, so the two $PreBias$ settings retain distinct transition rules in all simulated conditions.

Interaction mechanisms

The communication process in our model (see Figure 1) is inspired by Polya-urn (Gong et al., 2012). Every agent has a speech stack of length 3, which is filled with N (new variants) or O (old variants), to represent their language use. Initially, the speech stack of every node is set to [O O O], indicating exclusive use of the old form. One node is then selected and assigned a speech stack of [N N N], representing the introduction of an innovation. At each time step, every agent has one opportunity to initiate communication with its neighbors. At the beginning of every time step, a new randomized order is generated, and each agent acts once in that order. Speech-stack changes are applied immediately after each interaction rather than synchronously at the end of the time step. Consequently, agents acting later in the same time step respond to already updated speech stacks. In the one-to-many scenario, recipients are also processed in a randomized order. Each agent then selects a random variant from its speech stack and tries to spread it to its neighbors. If successful, the recipient replaces an opposing variant in its speech stack with that form. Notably, both variants are spreadable.

Communication scenarios

Language change encompasses a wide range of scenarios. The current study focuses on a specific scenario centered around two distinct modes of communication (one-to-one and one-to-many) that underlie various contexts. For instance, Rogers (2003) noted distinct dynamics in mass-media-like versus interpersonal communication for innovation spread, while Ananthasubramaniam et al. (2024) studied language diffusion differences between rural and urban areas. A common thread across these examples is variation in how many interlocutors may be reached in a single communicative act: some settings involve contact with multiple neighbors, whereas others are closer to dyadic exchanges between a speaker and a listener. Despite this, few studies have explored the impact of such differences on language change (except for Lee, 2024). Moreover, these distinctions offer a richer landscape for investigating centrality effects. To capture these two scenarios, our model runs as

Table 2. Parameter settings for simulation.

Parameter	Symbol	Values
population	n	500
proportion of agents preferring overt prestige	$PropAO$	0 0.2 0.4 0.6
prestige bias	$PreBias$	10 20
basic adoption rate	r	0.1
centrality		targeted log-bin sampling
scenario		one-to-one; one-to-many

follows: at each time step, a node initiates communication either with a single randomly selected neighbor or with all neighbors simultaneously (see Figure 1). Thus, the one-to-many scenario should be understood as a local one-to-many interaction rule rather than a strictly one-directional broadcasting process: agents can influence all neighbors when they act, but they can also be influenced by those neighbors when the neighbors act.

Simulation setup

We set up sixteen sets of simulations ($PropAO \times PreBias \times scenario$). Each set was run 1,000 times. To assign the initial innovator, each generated network was divided into ten logarithmically spaced degree bins, target bins were assigned cyclically across runs, and an innovator was randomly sampled from the targeted bin. Because the maximum degree varied across finite networks and nodes near that maximum were rare, the realized centrality counts were not balanced when the sampled degrees were mapped to common bins across all simulations. In particular, the 10th common bin contained only 86 of the 16,000 runs (0.54%), with a median of six runs per parameter-by-scenario condition. Figure 2 reports the realized sampling distribution. Here, “initiated by” refers to the assigned starting point of diffusion in the simulation, not to the probability that an individual generates an innovation in real-world language use. In all settings, the basic adoption rate r is fixed to 0.1, $PropAO$ ranges from 0 to 0.6, and $PreBias$ ranges from 10 to 20. These settings were selected because they allowed successful diffusion to occur while retaining sufficient variation across outcomes. Combined with centrality and scenario, the model parameters are presented in Table 2.

Equation-based models

Consistent with other diffusions, sigma curves are employed to characterize the dynamics of language diffusion. For instance, Piotrovskaja and Piotrovskij (1974) used an arctangent function to fit empirical data well. Shen (1990) and Weinreich et al. (1968) proposed logistic models to describe sound change. Based on them, Altmann (1992) derived three possible variants of the pattern of language change, collectively known as the Piotrovski-Altman law. This law declared that everything in language changes as a result of the interaction between old forms and new forms, which can be presented by Equation 1:

$$p(t) = \frac{C}{1 + e^{-bt+ct^2}}, \quad (1)$$

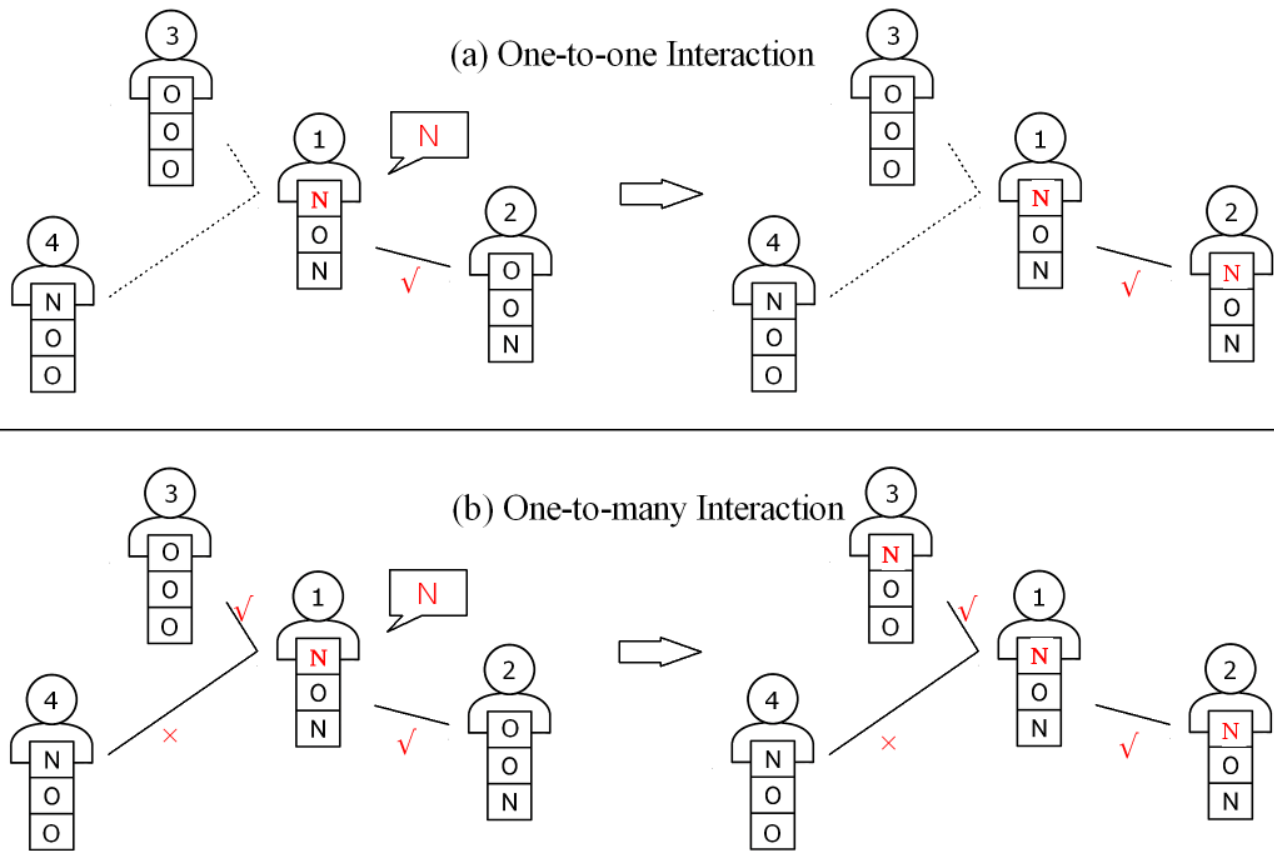


Figure 1 Interactions between agents in (a) the one-to-one scenario and (b) the one-to-many scenario. Speech stacks are labeled by letters on agents, and social connections are shown as lines. In the one-to-one scenario, Agent 1 selects an N variant from its speech stack and spreads it to Agent 2, who adopts it and changes an O to N in their own stack. In the one-to-many scenario, Agent 1 tries to spread it to all its neighbors. Agents 2 and 3 adopt it and update their speech stacks, while Agent 4 refuses to do so.

where $p(t)$ is the proportion of new forms or their users at time t , C is the limit of change, and b is a parameter controlling the diffusion intensity. This model has been widely applied to model language change (Gorski and Eder, 2023).

As for language diffusion, the partial variant might be more adaptive. For a clearer interpretation of curve-shape parameters, Stachowski (2013) transformed it into Equation 2:

$$p(t) = \frac{C}{1 + e^{-b(t-A)}}, \quad (2)$$

where $A = \ln(a)/b$. Parameter A marks the turning point of the influx rate, b reflects the intensity of language change, and C indicates strength (the upper limit of the population adopting the variant).

Compared to other mathematical models, such as the Abrams-Strogatz model (Abrams and Strogatz, 2003), Equation 2 is well suited to summarizing the shapes of diffusion curves. In the present analysis, Equation 2 was fitted to the simulated diffusion trajectories by nonlinear least squares with bounded parameter estimates. The equation was used as a descriptive summary of the simulated trajectories rather than as an independent generative model. Because the function assumes a monotonic trajectory approaching an upper limit, goodness of fit was evaluated for each fitted curve before the curve-shape parameters were analysed.

Measures

Social centrality

Network science provides several centrality measures, such as degree, betweenness, closeness, and eigenvector centrality, each capturing a different aspect of a node's position in a network (Newman, 2010). In this study, social centrality was measured as the number of links or neighbors an individual possesses within the network. This measurement is grounded in two considerations. First, established scales for assessing an individual's connectivity within a social network incorporate the number of social ties (Milroy, 1987). Second, the number of neighbors often serves as a robust indicator of node centrality within the network (Barabási, 2016). In this study, we focus on degree centrality because it directly represents the number of social ties, is straightforward to compute, and provides the clearest theoretical link to our question about how the quantity of social connections shapes diffusion. By adopting this measurement, we effectively isolate the influence of link quantity on centrality, explicitly excluding factors such as weights or directionality, ensuring that observed effects are attributable solely to the number of social connections. To clarify the role of centrality at different stages, we examine its effect from both the innovator and adopter perspectives.

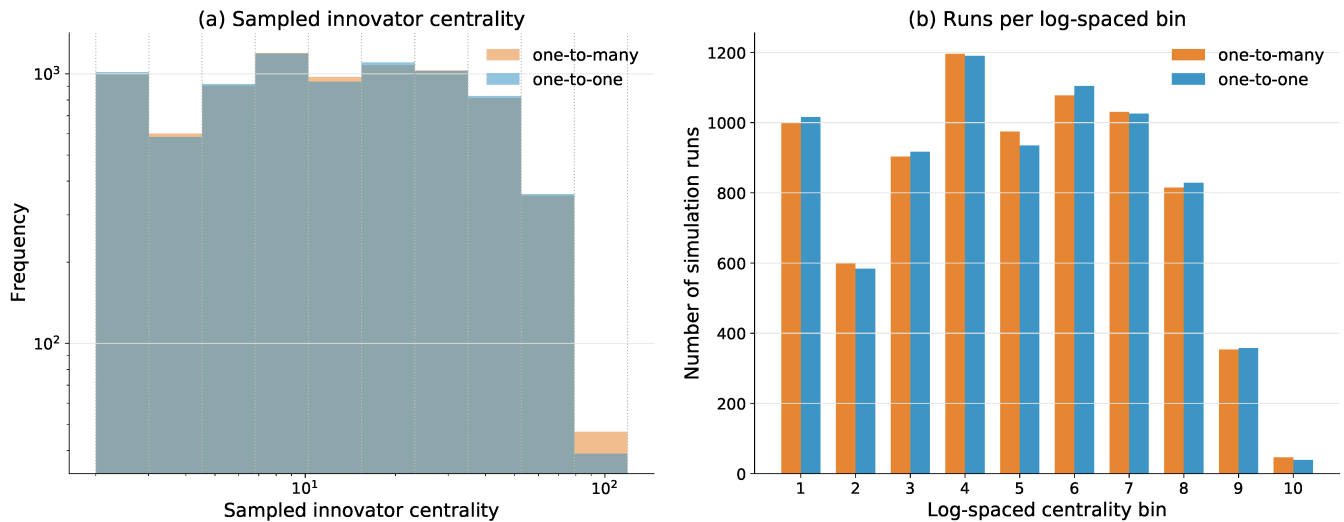


Figure 2 Realized sampling distribution of initial innovators. (a) Frequency of sampled innovator centrality values by communication scenario. Both axes are logarithmic, and vertical dotted lines indicate the common log-spaced centrality-bin boundaries. (b) Number of simulation runs falling within each realized centrality bin. Although target bins were assigned cyclically within each generated network, the realized counts were uneven after degrees were mapped to common bins across simulations, with especially sparse coverage in the 10th bin. The 10th bin is shown descriptively but excluded from all quantitative analyses.

Success rate

Measured as the probability that a new variant is adopted by at least 80% of the population under a given experimental setting (for example, $PropAO = 0$, $PreBias = 10$, and scenario = one-to-one), reflecting the overall effect of factors on diffusion potential.

Average adoption time

Measured as the average time steps a node requires to adopt a variant under specific parameters. A shorter adoption time indicates a node's tendency to adopt innovations early, highlighting its role in accelerating diffusion. In contrast, longer times may signal barriers.

Shape of diffusion curves

Measured as the inflection point, steepness, and height of the curves (fitted by Equation 2). These parameters capture the turning point, intensity, and strength of diffusion, respectively. To achieve smoother and more stable curve fitting, simulation trajectories were averaged within each combination of parameter condition, scenario, and logarithmically spaced innovator-centrality bin before fitting Equation 2.

Analysis

We used generalized linear mixed models (GLMMs) to examine the effects of centrality and communication scenario on the diffusion outcomes. GLMMs were used because the dependent variables differed in their measurement scales and distributional properties. Success rate was analysed as a binomial outcome, average adoption time as a positive continuous time variable, and curve-shape parameters as positive or bounded continuous variables. All models were fitted in *R* (Team., 2016) using the

glmmTMB package (Brooks et al., 2017). Innovator centrality and adopter centrality were mean-centered before analysis. Scenario was treatment-coded, with the one-to-many condition as the reference level. The experimental condition defined by $PropAO$ and $PreBias$ was included as a random intercept to account for variation across parameter settings.

All quantitative analyses excluded the 10th realized innovator-centrality bin. This bin represented only 0.54% of the simulation runs and contained too few realizations within individual parameter and scenario conditions to support representative estimates. Excluding it also prevented the highest, sparsely sampled centrality values from exerting disproportionate leverage on the fitted effects. The original 10th-bin runs were retained in the descriptive sampling distribution in Figure 2, but the success-rate, adoption-time, and curve-shape models were fitted using bins 1–9 only.

Three types of models were employed to examine the measures: (I) models assessing the effects of independent variables on success rate; (II) models comparing average adoption time across experimental conditions; and (III) models evaluating the influence of independent variables on the shape of diffusion curves. Figure 4 visualizes the model-based relationships between centrality and the three diffusion outcomes: success rate, average adoption time, and curve-shape parameters.

Models of type (I) examined how innovator centrality and scenario jointly influenced success rate. The dependent variable was specified as a grouped binomial response: for each combination of condition, innovator centrality, and scenario, we modelled the number of runs that met the threshold used to define success rate together with the number of runs that did not meet this threshold. Accordingly, each point in Figure 4(a) represents an aggregated success proportion, which may be 0, 1, or an intermediate value. We fitted a binomial GLMM with a logit link. Fixed effects included innovator centrality, scenario,

and their interaction term. The interaction term tested whether the effect of innovator centrality on success rate differed between communication scenarios.

$$\text{logit}(p_{\text{success}}) = \beta_0 + \beta_1 C_i + \beta_2 S + \beta_3 C_i \times S + u_{\text{condition}}.$$

Models of type (II) investigated factors influencing average adoption time. The dependent variable was the mean number of ticks required for adoption, calculated for each combination of condition, scenario, innovator centrality, and adopter centrality. Thus, Figure 4(b) and (c) use the same type of aggregated adoption-time observations, but display their relationship with innovator centrality and adopter centrality, respectively. Because average adoption time was positive and right-skewed, we fitted a Gamma GLMM with a log link. Fixed effects included innovator centrality, adopter centrality, scenario, and the interactions between scenario and each centrality term. Observations were weighted by the number of adoptions contributing to each mean. This model tested whether average adoption time depended on both the centrality of the innovator and the centrality of the adopting individual, and whether these effects varied between communication scenarios.

$$\begin{aligned} \log(\mu_T) = & \beta_0 + \beta_1 C_i + \beta_2 C_a + \beta_3 S \\ & + \beta_4 C_i \times S + \beta_5 C_a \times S + u_{\text{condition}}. \end{aligned}$$

Models of type (III) evaluated the influence of innovator centrality and scenario on the shape of diffusion curves. For each combination of condition, scenario, and logarithmically spaced innovator-centrality bin, repeated simulation runs were averaged and fitted with the partial Piotrovski-Altmann function in Equation 2. The representative innovator-centrality value for each bin was defined as the median sampled centrality within that bin. Thus, each point in Figure 4(d-f) represents a fitted parameter from an averaged trajectory, rather than an individual simulation run or time step. The resulting parameters described the inflection point (A), diffusion intensity (b), and diffusion strength (C) of the curve. Only curve fits with $R^2 > .75$ were included in the curve-shape analyses. After the 10th bin was excluded, parameters A and b were analysed using Gamma GLMMs with log links, whereas C , which is bounded between 0 and 1, was analysed using a beta GLMM with a logit link. Fixed effects included innovator centrality, scenario, and their interaction.

$$\begin{aligned} g(\mu_\theta) = & \beta_0 + \beta_1 C_i + \beta_2 S + \beta_3 C_i \times S + u_{\text{condition}}, \\ & \theta \in \{A, b, C\}. \end{aligned}$$

Here, C_i denotes innovator centrality, C_a denotes adopter centrality, S denotes communication scenario, and $u_{\text{condition}}$ denotes the random intercept for each parameter condition. For the curve-shape models, $g = \log$ for A and b , and $g = \text{logit}$ for C . In the Results, we report coefficient estimates, standard errors, test statistics, and p values; where appropriate, exponentiated coefficients $\exp(\beta)$ and their 95% confidence intervals are also reported to facilitate interpretation of effects on the response scale. Detailed model-fitting information and diagnostic summaries are reported in the Supplementary Materials, including the fitted curve parameters and R^2 values. Among the 144 averaged trajectories in bins 1–9, 143 met the criterion $R^2 > .75$.

Results

We present the results for each of the linguistic measures (success rate, average adoption time, and curve shapes) separately, based on three types of analyses (see Methods). Figure 3 exemplifies the diffusion curves under various settings, while Figure 4 illustrates the relationships between the measures and factors.

Success rate

Results revealed significant main effects of both innovator centrality and scenario on success rate (see Figure 4(a)). Innovator centrality had a significant negative effect on the odds of successful diffusion ($\beta = -0.070$, $SE = 0.002$, $z = -34.47$, $p < .001$; $\exp(\beta) = 0.933$, 95% CI [0.929, 0.936]), indicating that higher innovator centrality was associated with lower success rates. In contrast, the one-to-one scenario yielded significantly higher odds of success than the one-to-many scenario ($\beta = 0.609$, $SE = 0.066$, $z = 9.29$, $p < .001$; $\exp(\beta) = 1.839$, 95% CI [1.617, 2.091]). The interaction between innovator centrality and scenario was not statistically significant ($\beta = 0.004$, $SE = 0.003$, $z = 1.54$, $p = .124$; $\exp(\beta) = 1.004$, 95% CI [0.999, 1.010]), suggesting that the negative association between innovator centrality and success rate did not differ reliably across the two scenario conditions. Overall, although the one-to-one condition increased the likelihood of successful diffusion, higher innovator centrality was consistently associated with lower success rates.

Average adoption time

Results revealed a significant main effect of innovator centrality on average adoption time (see Figure 4(b-c)). In the one-to-many scenario, innovator centrality had a significant negative effect on average adoption time ($\beta = -0.006$, $SE < 0.001$, $z = -850.20$, $p < .001$; $\exp(\beta) = 0.994$, 95% CI [0.994, 0.994]), indicating that innovations initiated by more central individuals were adopted more rapidly. The interaction between innovator centrality and scenario was also significant ($\beta = 0.009$, $SE < 0.001$, $z = 1010.64$, $p < .001$; $\exp(\beta) = 1.009$, 95% CI [1.009, 1.009]), indicating that the effect of innovator centrality on average adoption time varied across scenarios. Specifically, in the one-to-one scenario, higher innovator centrality was associated with longer average adoption times.

Results also showed a significant main effect of adopter centrality ($\beta = -0.011$, $SE < 0.001$, $z = -686.89$, $p < .001$; $\exp(\beta) = 0.989$, 95% CI [0.989, 0.989]), indicating that more central adopters adopted the new variant earlier. The interaction between adopter centrality and scenario was significant ($\beta = -0.004$, $SE < 0.001$, $z = -203.15$, $p < .001$; $\exp(\beta) = 0.996$, 95% CI [0.995, 0.996]), suggesting that the negative effect of adopter centrality on average adoption time was stronger in the one-to-one scenario.

Finally, the main effect of scenario was significant. Compared with the one-to-many scenario, the one-to-one scenario was associated with significantly longer average adoption times ($\beta = 1.606$, $SE < 0.001$, $z = 2852.38$, $p < .001$; $\exp(\beta) = 4.984$, 95% CI [4.978, 4.989]), indicating that the one-to-many scenario

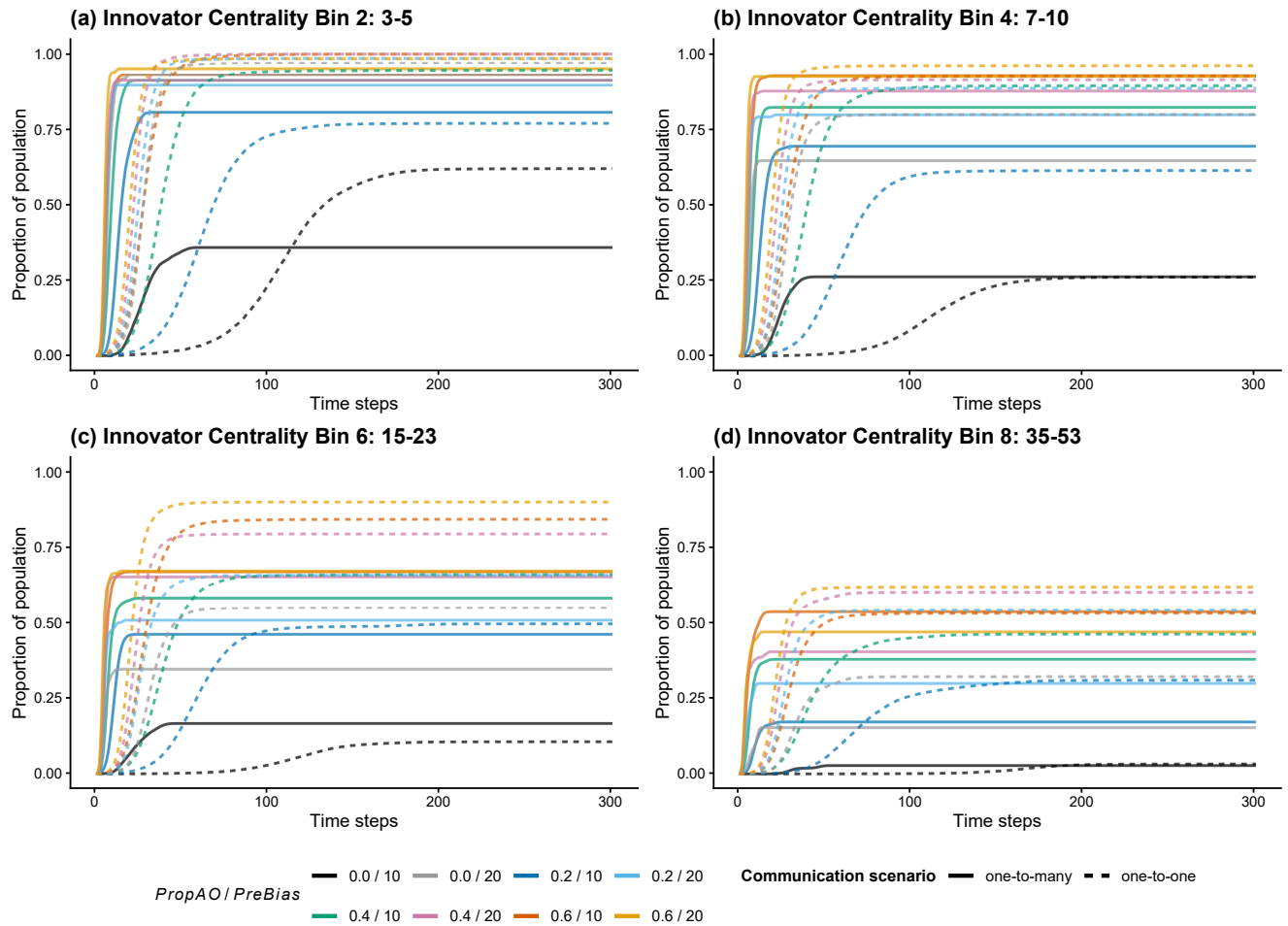


Figure 3 Representative mean diffusion trajectories from the source simulation data under different *PropAO* and *PreBias* parameter combinations, shown for the 2nd, 4th, 6th, and 8th log-spaced innovator-centrality bins. The title of each panel reports the corresponding centrality range, rounded to integer values. These bins were selected at approximately equal intervals across the retained centrality range. Colors identify the eight *PropAO/PreBias* combinations listed in the legend, with each label giving *PropAO* followed by *PreBias*. Communication scenarios are distinguished by line type: one-to-many is shown with solid lines and one-to-one with dashed lines.

generally facilitated faster adoption, whereas diffusion proceeded more slowly in the one-to-one scenario.

Overall, average adoption time was jointly shaped by innovator centrality, adopter centrality, and communication scenario. More central adopters adopted earlier, whereas the effect of innovator centrality depended on the scenario: it shortened adoption time in the one-to-many scenario but lengthened adoption time in the one-to-one scenario.

Curve shape A: inflection points

Results revealed a significant main effect of innovator centrality on curve-shape parameter *A* ($\beta = -0.0027$, $SE < 0.001$, $z = -4.63$, $p < .001$; $\exp(\beta) = 0.9973$, 95% CI [0.9961, 0.9984]), indicating that, in the one-to-many scenario, higher innovator centrality was associated with earlier inflection points.

The interaction between innovator centrality and scenario was significant ($\beta = 0.0058$, $SE < 0.001$, $z = 7.06$, $p < .001$; $\exp(\beta) = 1.0059$, 95% CI [1.0042, 1.0075]), indicating that the effect of innovator centrality on *A* varied across

scenarios. Specifically, in the one-to-one scenario, higher innovator centrality was associated with later inflection points.

The main effect of scenario was also significant. Compared with the one-to-many scenario, the one-to-one scenario was associated with significantly later inflection points ($\beta = 1.486$, $SE = 0.015$, $z = 96.67$, $p < .001$; $\exp(\beta) = 4.418$, 95% CI [4.287, 4.553]), indicating that diffusion in the one-to-one scenario generally reached the deceleration phase later, whereas the turning point emerged earlier in the one-to-many scenario.

Overall, parameter *A* was shaped by both innovator centrality and communication scenario. The effect of innovator centrality depended on the scenario: higher innovator centrality was associated with earlier inflection points in the one-to-many scenario but later inflection points in the one-to-one scenario. In addition, the one-to-one scenario was generally characterized by later diffusion inflection points.

Curve shape b: intensity

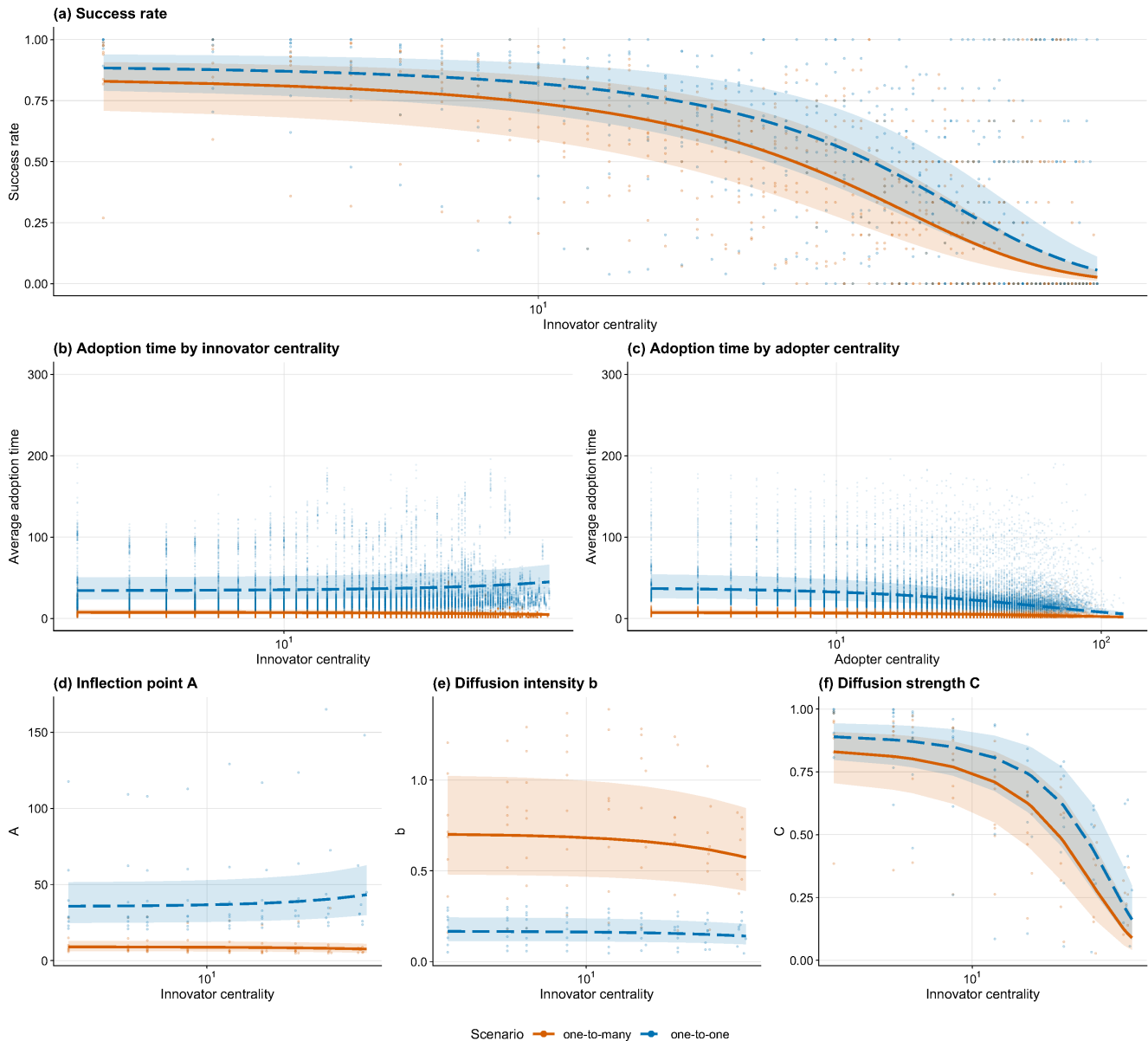


Figure 4 Changes in (a) success rates, (b)–(c) average adoption time, and (d)–(f) GLMM-estimated curve-shape parameters as a function of innovator/adopter centrality under two scenarios. Centrality values are shown on log-scaled x-axes. Data from the one-to-many and one-to-one scenarios are shown in orange solid lines and blue dashed lines, respectively. All quantitative estimates exclude the sparsely sampled 10th innovator-centrality bin. Curve-shape estimates are based on trajectories averaged within retained logarithmically spaced innovator-centrality bins and fitted with Equation 2.

Results revealed a significant main effect of innovator centrality on curve-shape parameter b ($\beta = -0.0032$, $SE = 0.0010$, $z = -3.16$, $p = .002$; $\exp(\beta) = 0.9968$, 95% CI [0.9948, 0.9988]), indicating that higher innovator centrality was associated with lower diffusion intensity in the one-to-many scenario.

The interaction between innovator centrality and scenario was not statistically significant ($\beta = 0.0005$, $SE = 0.0014$, $z = 0.36$, $p = .722$; $\exp(\beta) = 1.0005$, 95% CI [0.9978, 1.0032]), indicating that the negative association between innovator centrality and b did not differ reliably across the two scenarios.

The main effect of scenario was also significant. Compared with the one-to-many scenario, the one-to-one scenario was

associated with significantly lower b values ($\beta = -1.428$, $SE = 0.025$, $z = -56.23$, $p < .001$; $\exp(\beta) = 0.240$, 95% CI [0.228, 0.252]), indicating that the one-to-many scenario generally showed greater diffusion intensity, whereas diffusion intensity was lower in the one-to-one scenario.

Overall, parameter b was shaped by innovator centrality and communication scenario. Higher innovator centrality was associated with lower diffusion intensity, with no reliable difference in this association across scenarios. In addition, the one-to-many scenario was generally associated with greater diffusion intensity.

Curve shape *C*: strength

Results revealed a significant main effect of innovator centrality on curve-shape parameter *C* ($\beta = -0.063$, $SE = 0.004$, $z = -14.30$, $p < .001$; $\exp(\beta) = 0.939$, 95% CI [0.930, 0.947]), indicating that, in the one-to-many scenario, higher innovator centrality was associated with smaller *C* values. In other words, innovations initiated by more central individuals reached lower final levels of diffusion.

The main effect of scenario was also significant. Compared with the one-to-many scenario, the one-to-one scenario was associated with significantly higher *C* values ($\beta = 0.557$, $SE = 0.101$, $z = 5.54$, $p < .001$; $\exp(\beta) = 1.746$, 95% CI [1.433, 2.126]), indicating that the one-to-one scenario generally corresponded to higher final levels of diffusion.

The interaction between innovator centrality and scenario was not statistically significant ($\beta = 0.003$, $SE = 0.006$, $z = 0.51$, $p = .612$; $\exp(\beta) = 1.003$, 95% CI [0.992, 1.014]), indicating that the negative effect of innovator centrality on *C* did not differ reliably across the two scenarios. That is, although the one-to-one scenario was generally associated with higher *C* values, the tendency for higher innovator centrality to correspond to lower final diffusion levels was broadly consistent across both scenarios.

Overall, parameter *C* was mainly shaped by innovator centrality and communication scenario. More central innovators were associated with lower final levels of diffusion, whereas the one-to-one scenario was generally associated with higher final levels of diffusion; however, there was no significant interaction between the two.

Discussion

We employed a computational simulation paradigm to examine how social centrality and communication scenario jointly influence the dynamics of linguistic diffusion. We expected individuals with different levels of centrality to contribute differently to the diffusion process, while distinguishing between the assigned source of an innovation and the subsequent propagation of that innovation. Moreover, these roles were expected to be moderated by the scenario in which communication occurred. Our results showed that, conditional on an individual being selected as the initial innovator in the simulation, linguistic variants originating from less central individuals had a higher diffusion success rate than those introduced by more central individuals. Notably, more central adopters were found to adopt new variants earlier than their less central counterparts, whereas the timing effect of innovator centrality depended on the communication scenario. Furthermore, compared with variants introduced by low-centrality individuals, those originating from highly central ones showed lower diffusion intensity and lower final diffusion levels. Finally, the effects of communication scenario differed across diffusion measures: one-to-one scenarios were associated with higher success rates and final diffusion levels but slower adoption and lower diffusion intensity, whereas one-to-many scenarios were associated with faster adoption and stronger diffusion intensity. Together, the findings indicate that social centrality, moderated by the communication scenario, jointly shapes the diffusion process of linguistic variants.

Simulations only show what could happen, not necessarily what must have happened (Gong et al., 2012; Hurford, 2012). Nevertheless, the findings may offer insights into real-world communities, as social centrality and communication scenarios are highly relevant to patterns of interaction observed in natural social networks. Moreover, our results not only validate the necessity of individual differentiation for language change (Gong et al., 2008; Blythe and Croft, 2012; Baxter, 2016) but also remain consistent with other specific computational simulation findings (Fagyal et al., 2010; Swarup et al., 2011). This connection should therefore be interpreted narrowly: low-centrality starting points were more favorable for successful diffusion, but the model does not show that low-centrality individuals were more likely to generate innovations. High-centrality individuals instead played a more limited role in propagation, especially through earlier adoption by central adopters, without increasing diffusion success, intensity, or final diffusion levels. This pattern also echoes empirical findings from social network research (Labov, 1972; Cheshire, 1982; Milroy, 1987; Eckert, 2000). Individuals with lower centrality show greater resistance to established norms, while those with higher centrality are more likely to maintain and reinforce norms. Finally, the results align with innovation diffusion studies (Rogers, 2003), which suggest that a one-to-many-like scenario, in which one actor can reach multiple recipients in a given interaction, spreads information more rapidly but within a narrower population, compared to a one-to-one-like scenario (e.g., interpersonal communications).

Our measures collectively capture the multifaceted influence of social centrality on linguistic diffusion. The complex effects of centrality on success rate, average adoption time, and curve shape reveal its essence, which reflects the coupling between the weak-tie model and the two-step flow model. Specifically, the success rate and final diffusion level reflect diffusion outcomes conditional on the assigned source of innovation. Effects on these measures suggest that innovations assigned to peripheral starting points are more likely to survive and spread successfully, rather than that peripheral individuals are more likely to generate innovations. In contrast, the shapes of diffusion curves and average adoption time distinguish source and adopter roles: more central adopters tend to adopt innovations earlier, whereas innovations introduced by more central individuals show lower diffusion intensity and scenario-dependent inflection timing. This coupling arises because individuals with different levels of centrality exhibit distinct responses to network norms (Milroy, 1987; Milroy and Milroy, 1992; Milroy and Llamas, 2013). In comparison with earlier work, our results specify this mechanism more precisely as a direct association between the number of social relations and diffusion measures, rather than other attributes (e.g., weight and type). Individuals with rich social connections serve as critical nodes in the competition between old and new variants. They can disseminate variants to a broader audience and, as adopters, encounter innovations earlier, thereby facilitating the rapid propagation of emerging forms. Hence, when an innovation originates from the network center, it faces greater resistance from adherents of existing variants in the early stages of diffusion. Conversely, variants that emerge from the network periphery, where individuals maintain fewer connections, encounter less early resistance and are consequently more likely to survive and spread successfully.

Another important aspect of our results concerns the effect of the communication scenario on diffusion processes. We found that, compared to one-to-many scenarios, one-to-one settings exhibit higher diffusion success rates and C values, but are associated with later inflection points, lower b values, and longer average adoption times for new variants. These results highlight the underlying mechanistic differences between one-to-one and one-to-many scenarios across various communication contexts (Balbi and Kittler, 2016). These differences help explain, from a more general perspective, the variations in diffusion observed across different contexts—such as rural versus urban settings (see their different patterns in Ananthasubramaniam et al., 2024). Importantly, this effect can also be explained by the number of social connections activated: one-to-many scenarios activate more social connections than one-to-one scenarios within the same time frame, which intensifies competition between new and old variants and may help explain the observed differences.

Our findings highlight the significance of centrality differentiation within social networks as a crucial factor of language diffusion. Specifically, we demonstrate that the spread of linguistic variants can be understood through the lens of individual social centrality and its scenario-dependent activation. Innovations assigned to more central starting points tend to show lower diffusion intensity and lower final reach, while their inflection timing depends on the communication scenario. Conversely, innovations assigned to lower-centrality starting points are more likely to become successful diffusion events and to reach higher final diffusion levels, possibly because peripheral positions are less constrained by early pressure from neighbors using the old variant. Although communication scenarios moderate the effects of social centrality, the functional differentiation of centrality remains a fundamental mechanism shaping language change. Our results refine the understanding of centrality effects by identifying their distinct contributions to the initiation and propagation phases of linguistic diffusion.

Our study underscores the crucial role of social network structure and interactional mechanisms in shaping the dynamics of language diffusion. From an ego star perspective, previous research has often conflated distinct aspects of individual centrality, overlooking the multifaceted ways in which network centrality influences linguistic diffusion (Fagyal et al., 2010; Milroy, 1987). Beyond the number of social connections, factors such as their weight, type, and directionality substantially modify an individual's effective centrality and thus their potential influence on linguistic change. Moreover, individual-level network parameters beyond degree centrality, such as other centrality measures, clustering coefficient, and average path length, have been shown to exert significant effects on diffusion processes (Josserand et al., 2024). The present study used Barabasi-Albert networks as a simplified scale-free topology, but did not systematically test whether the same results hold under alternative topologies such as small-world networks or networks with explicit community structure. Therefore, the generality of the findings across network structures should be examined in future work. In addition, centrality and exposure opportunities are not fully separable in the present model: agents with higher degree may encounter more variants and transmit them to more individuals within the same time step, while one-to-many interactions also involve more interaction events than one-to-one

interactions. Thus, differences in adoption time and diffusion speed may partly reflect differences in exposure opportunities rather than sociolinguistic mechanisms alone. Future research should aim to disentangle these interacting dimensions, for example by expressing adoption time relative to the number of interaction events or by comparing scenarios under controlled exposure opportunities, to clarify their specific contributions to the initiation, propagation, and stabilization of linguistic variants within evolving social systems.

Data availability

The data and code supporting this study are available at <https://github.com/YuhuYang/Supplementary-Materials-for-Social-Centrality-Differentiation>.

Conflicts of interest

The authors declare no conflicts of interest.

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